Systemadministration

Linux

Linux Command-Line

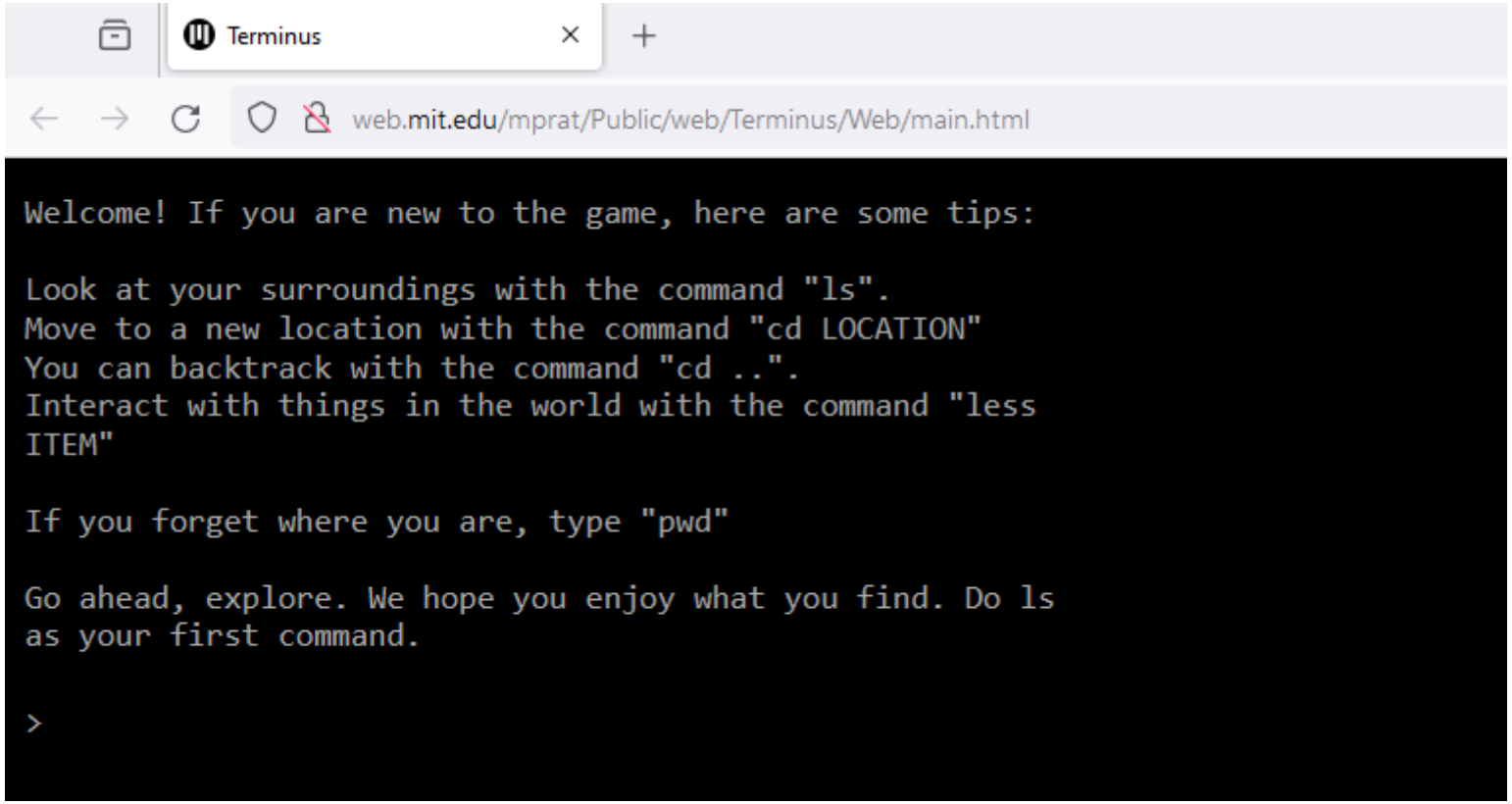
Agenda

Hur man använder Linux

1. Choose Your Own Adventure
2. The Linux Shell
3. File System Hierarchy
4. Navigering
5. Relativsökväg
6. Hantering av filer & kataloger
7. Användare & filrättigheter
8. Tips

Choose Your Own Adventure

MIT Terminus



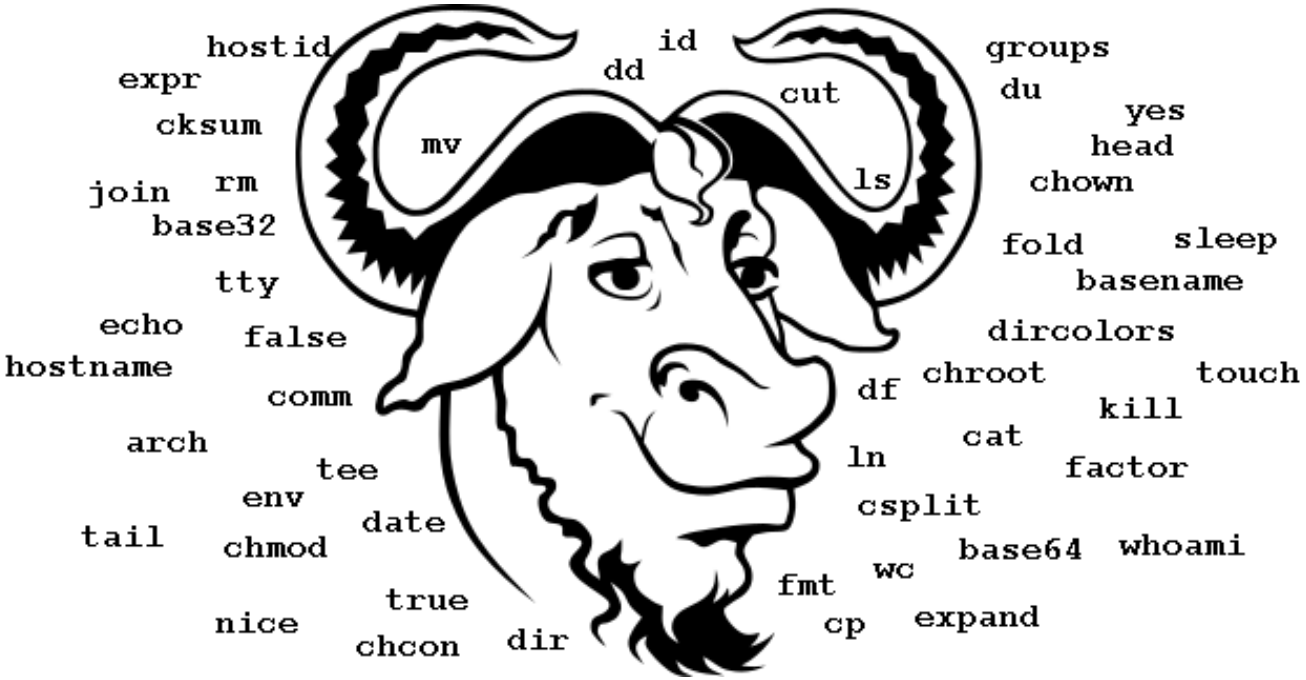
http://web.mit.edu/mprat/Public/web/Terminus/Web/main.html

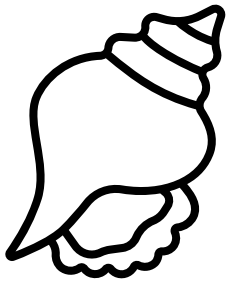
The Linux Shell

BASH
THE BOURNE-AGAIN SHELL

GNU BASH är det vanligaste **skalet**

GNU Coreutils ger UNIX-liknande CLI-program





Shell Prompt

Typiskt:

hostname

```
nics@linux:~$
```

username

```
nics@linux:/etc/$
```

path

Using Commands

```
command [--flag] [argument…]
command [-f] [argument…]
```

Ett kommando kan ta en/flera argument

Options anges med ’flaggor’

Long-form, short-form flags

Case sensitive!

Using Commands

Ex.

flag

```
nics@linux:~$ ls -l /etc/
```

command

argument

argument 1

```
nics@linux:~$ ls -l /etc/ /var/log/
```

argument 2

Using Commands

Flaggor kan kombineras

Ordningen spelar ingen roll

Ex.

```
nics@linux:~$ ls -l -h -F /etc/
```

```
nics@linux:~$ ls -lhF /etc/
```

Med ett undantag…

Using Commands

Förutom när en flagga tar ett argument

Ex.

```
nics@linux:~$ ssh-keygen -lvE sha256 -f /etc/…
```

-E Tar ett argument

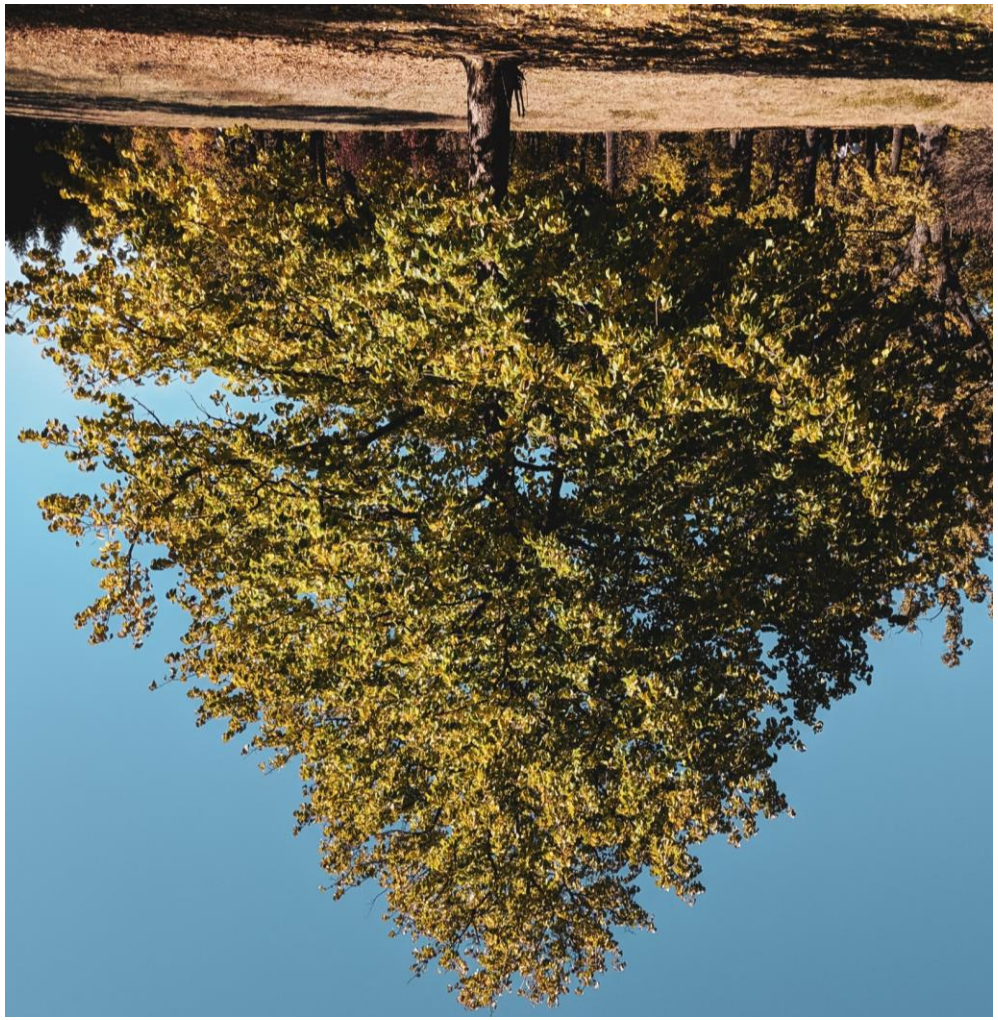
Antingen sha256 eller md5

```
ssh-keygen -Evl sha256
```

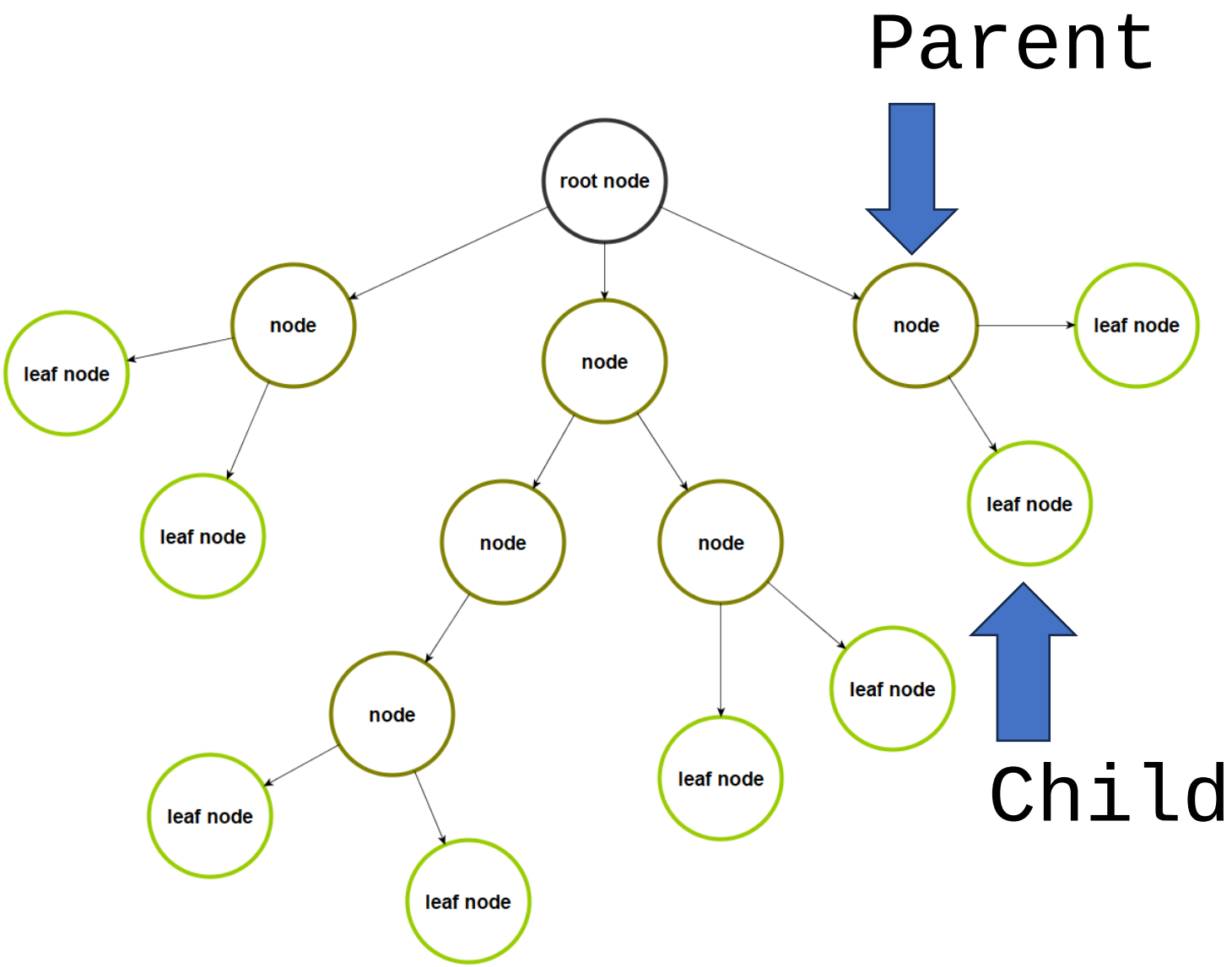
Inkorrekt!

File System Hierarchy

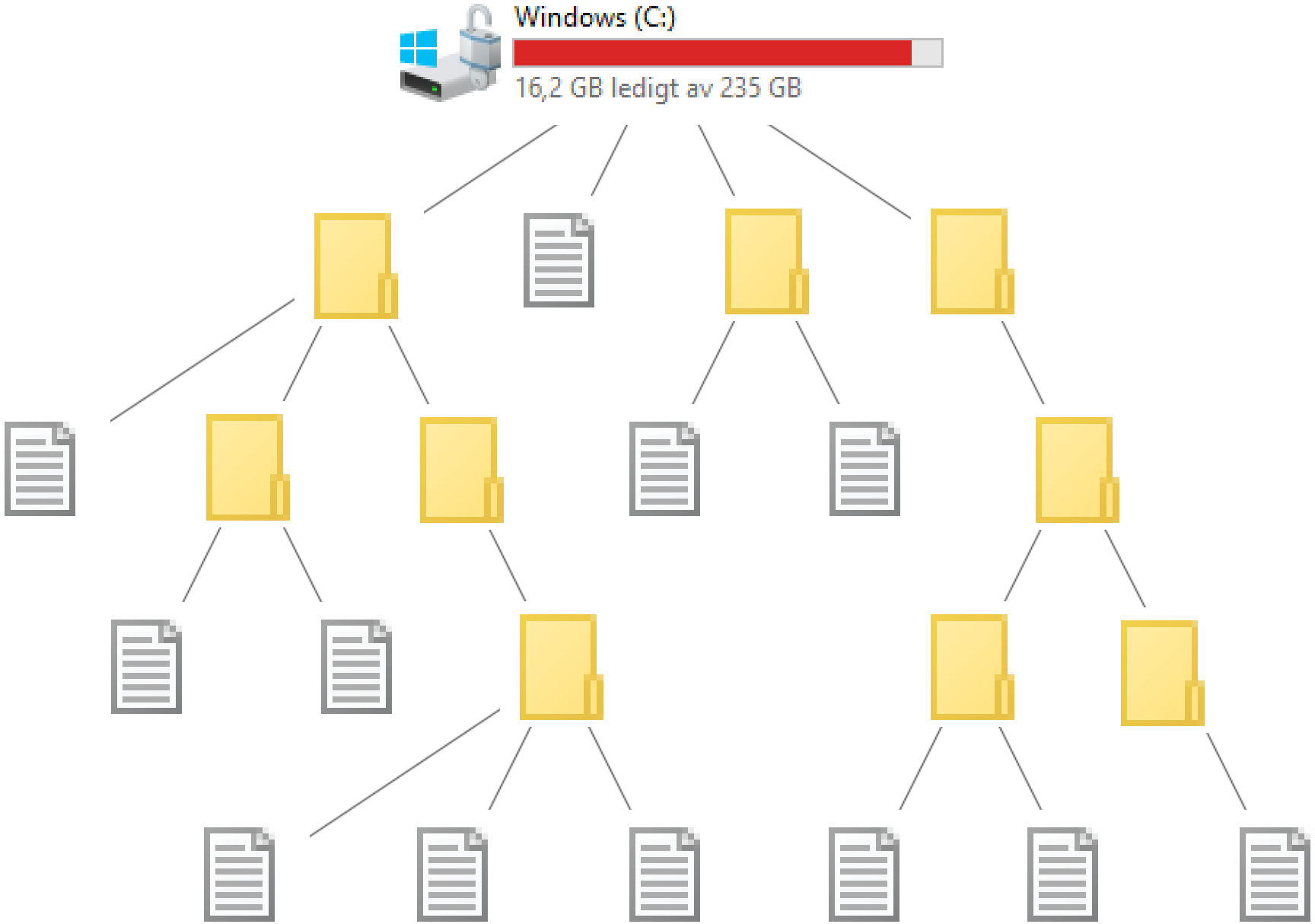
Filträd



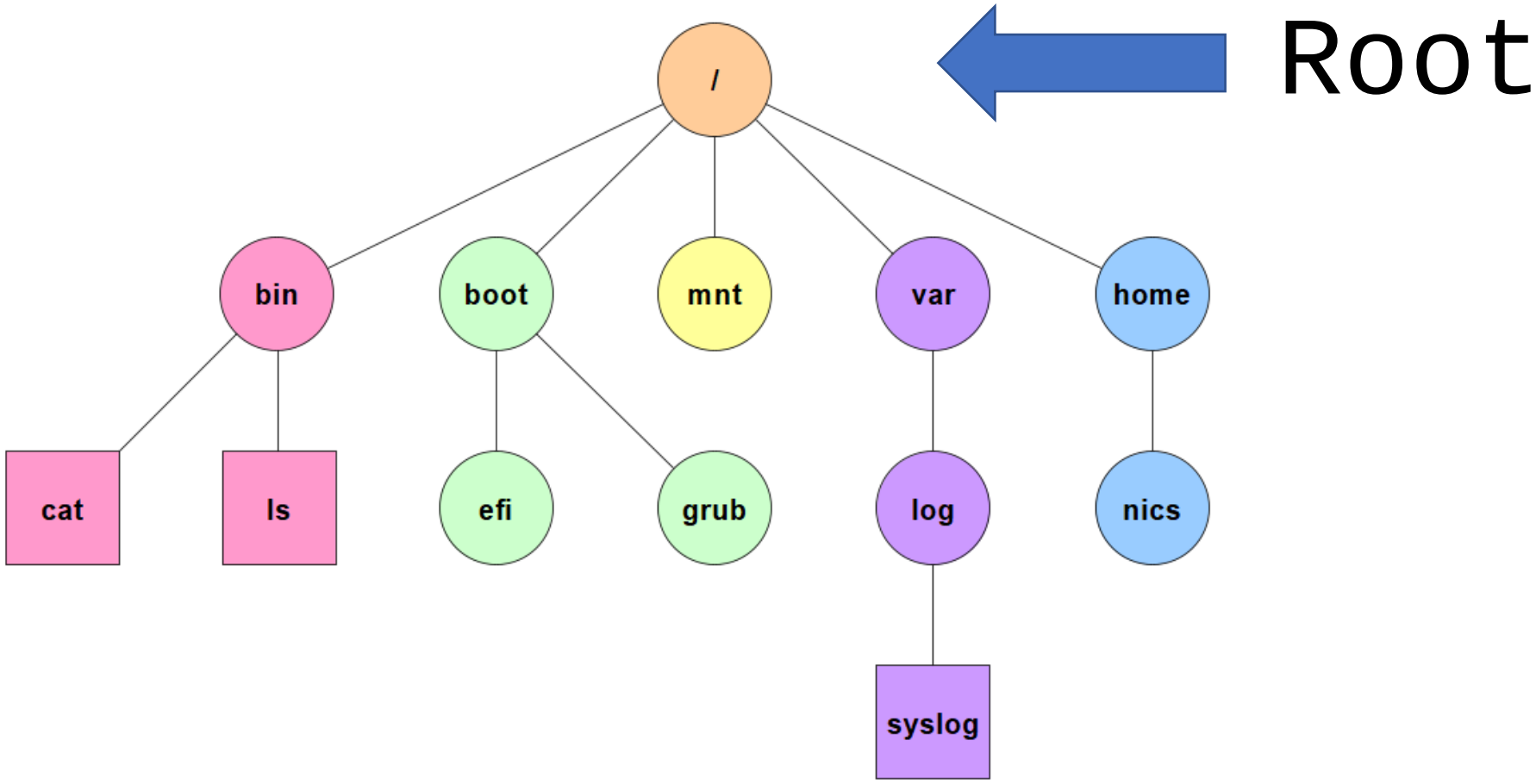
Green and yellow tree under blue sky during daytime, by Ivan Rohovchenko, 2019, Unsplash (https://unsplash.com/photos/p3vrfHDr46s). Unsplash license.



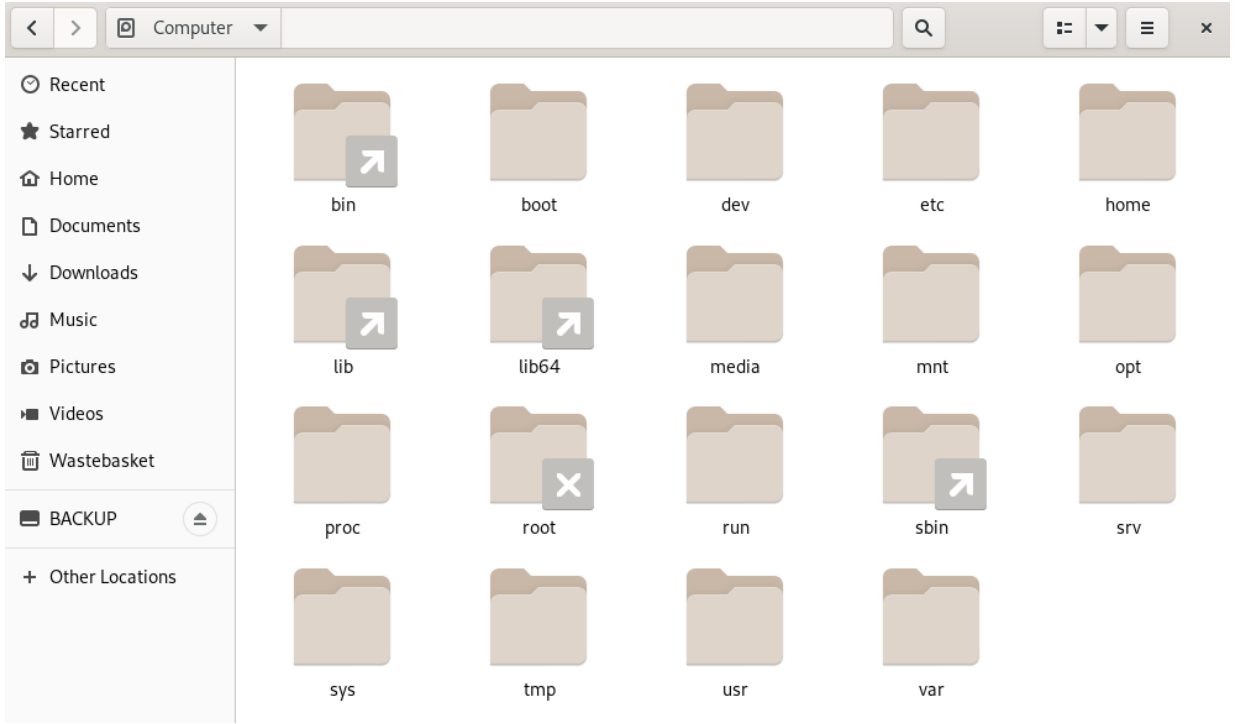
Filträd

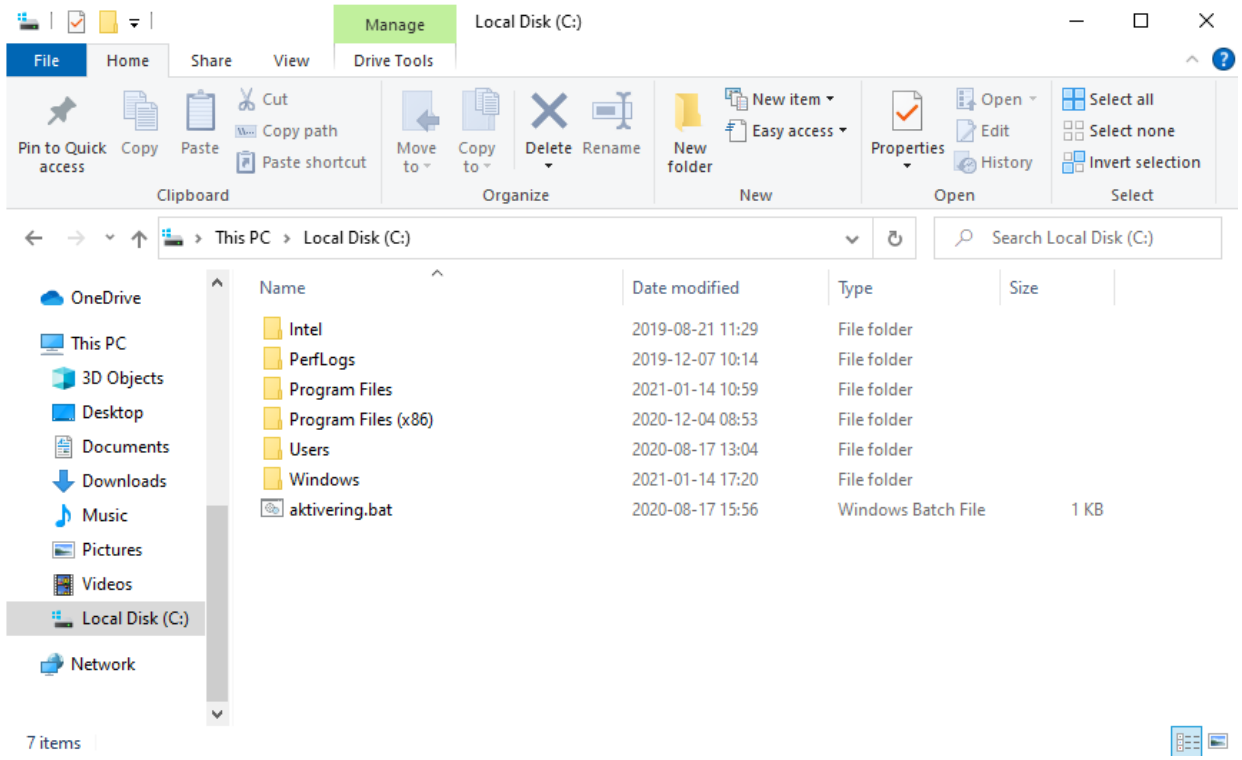


Filträd



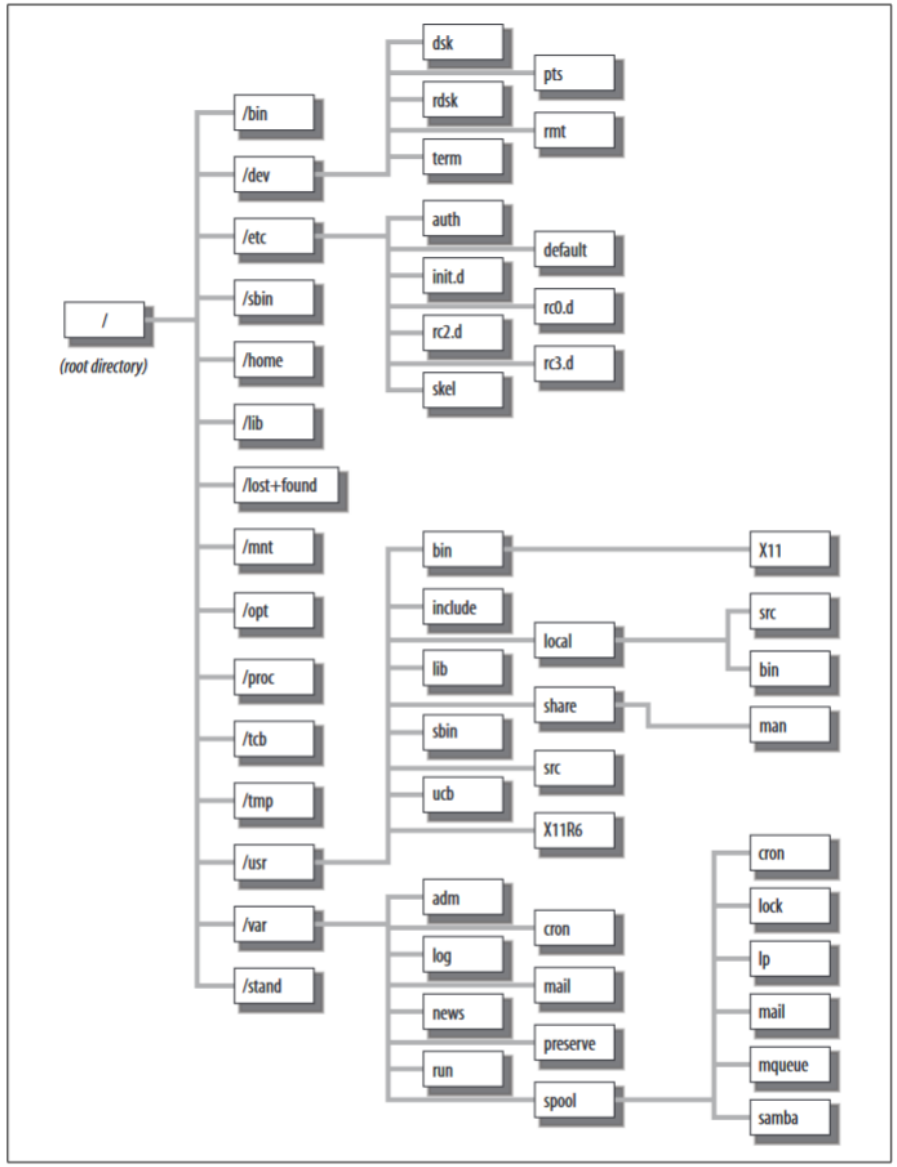
Filhierarki





File Hierarchy Standard [1994, 2004, 2015]

UNIX-liknande OS följer nästan alltid **FHS**



Generic Unix directory structure, pp. 69, by Æleen Frisch, 2002, O'Reily. Fair use/I enlighet med citaträtt

https://refspecs.linuxfoundation.org/FHS_3.0/fhs/index.html

/	*root directory*
/bin	*essential command executables (**bin**aries) for all users*
/boot	*programs & files used during **boot** (i.e. boot loader)*
/dev	***dev**ice files*
/etc	*system configuration files and scripts (**et c**etera)*
/home	*users' **home** directories*
/mnt	*temporarily **m**ou**nt**ed filesystems*
/media	*mount points for removable **media***
/sbin	*essential **s**ystem (or **s**uperuser) **bin**aries*
/usr	**us**e**r**-land programs & read-only data, shared by all users
/var	***var**iable data, such as logs, databases and websites*

/bin /usr vad är skillnaden?

/bin *essential command executables (**bin**aries) for all users*

/usr **user**-land programs & read-only data, shared by all users

WIKI

debian

UsrMerge

The Debian /usr Merge

This page tracks Debian support for the merged /usr directories scheme, i.e. the `/{bin,sbin,lib}/` directories becoming symbolic links to `/usr/{bin,sbin,lib}/`.

https://wiki.debian.org/UsrMerge

freedesktop.org

The Case for the /usr Merge

What's Being Discussed Here?

Fedora (and other distributions) have finished work on getting rid of the separation of /bin and /usr/bin, as well as /sbin and /usr/sbin, /lib and /usr/lib, and /lib64 and /usr/lib64. All files from the directories in / will be merged into their respective counterparts in /usr, and symlinks for the old directories will be created instead:

```
/bin → /usr/bin
/sbin → /usr/sbin
/lib → /usr/lib
/lib64 → /usr/lib64
```

https://www.freedesktop.org/wiki/Software/systemd/TheCaseForTheUsrMerge/

/bin /usr vad är skillnaden?

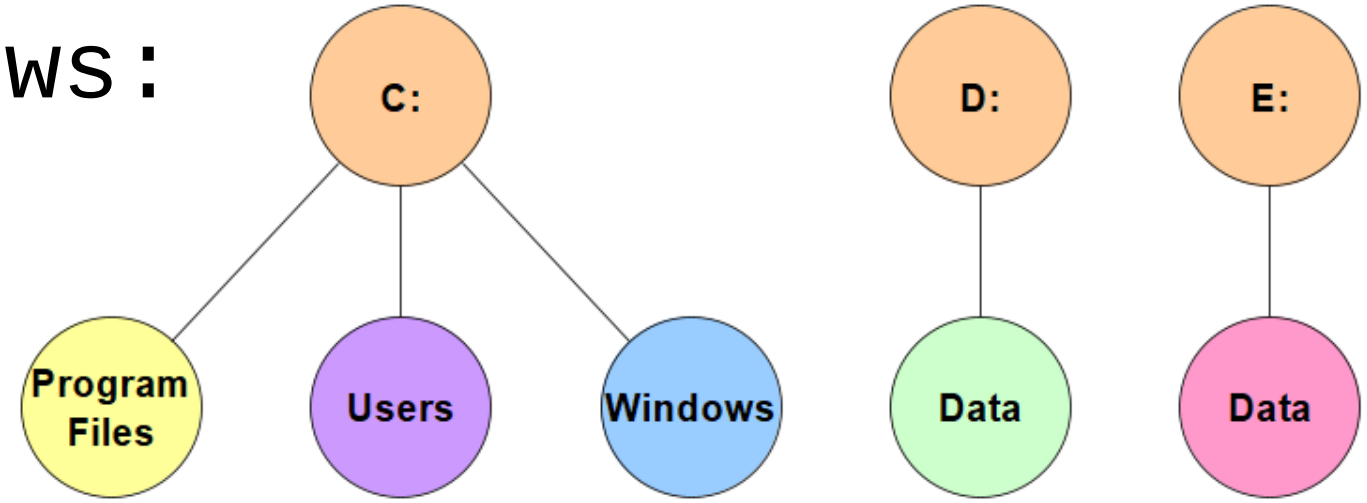
/bin *essential command executables (**bin**aries) for all users*

/usr **us**er-land programs & read-only data, shared by all users

```
nics@linux:~$ ls -ld /bin /usr
lrwxrwxrwx   root root    bin -> usr/bin
drwxr-xr-x   root root    usr
nics@linux:~$ ls /usr
bin
```

Multiple Disks (Mount Points)

Windows:

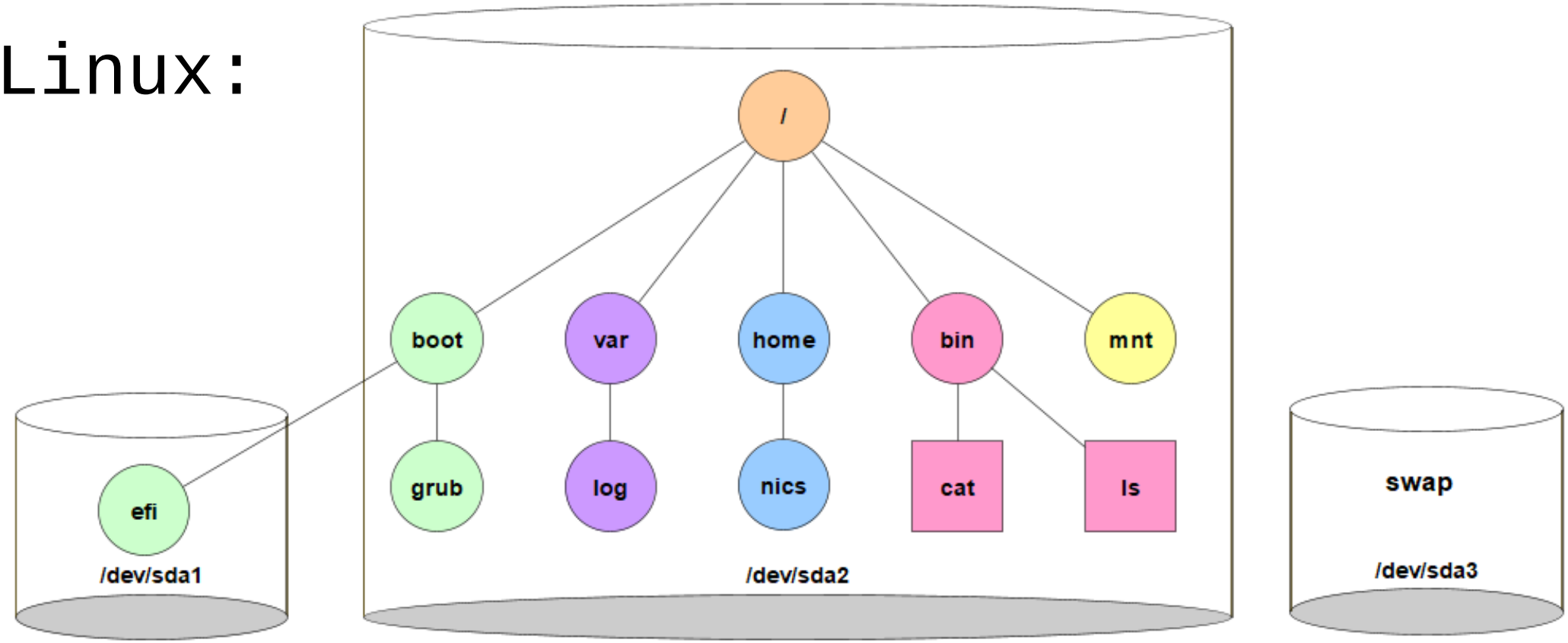


C: har en fördefinierad filstruktur

Ytterligare partitioner definieras av användaren

Multiple Disks (Mount Points)

Linux:



Filstrukturen är frikopplad från partitioner

Navigering

Filsökväg

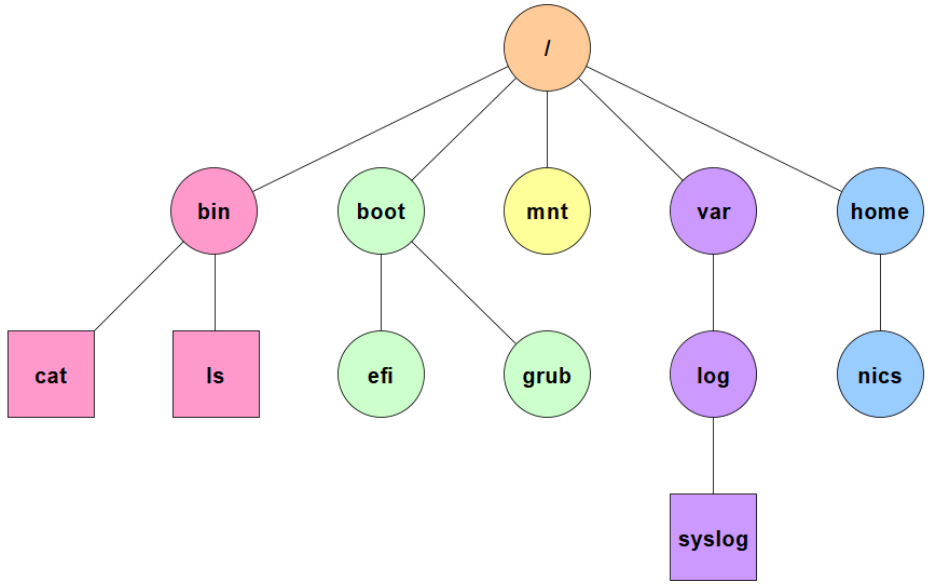
Path

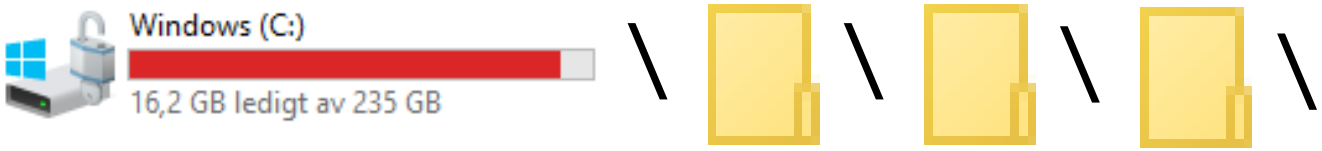
Ex. Linux

/home/nics/dir2/

Windows

C:\Users\nics\dir2\





Print Working Directory

För att visa nuvarande filsökväg

```
nics@linux:~/dir2$ pwd
/home/nics/dir2
nics@linux:~/dir2$
```

~

Tilde

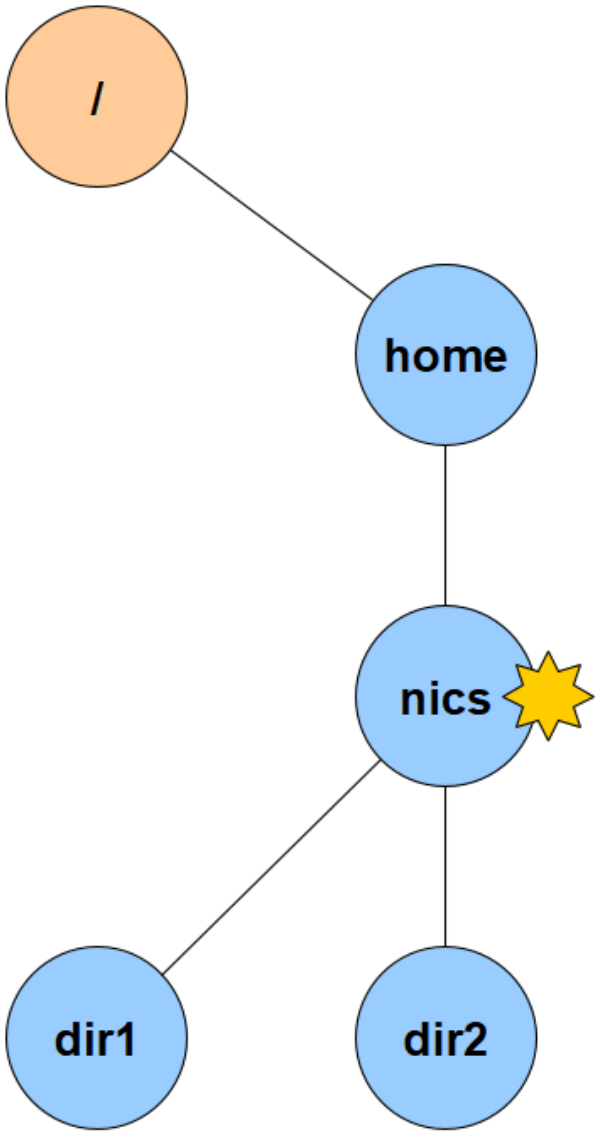
~ = din användares hemkatalog

```
nics@linux:~$ echo ~
/home/nics
```

```
nics@linux:~$ echo "$HOME"
/home/nics
```

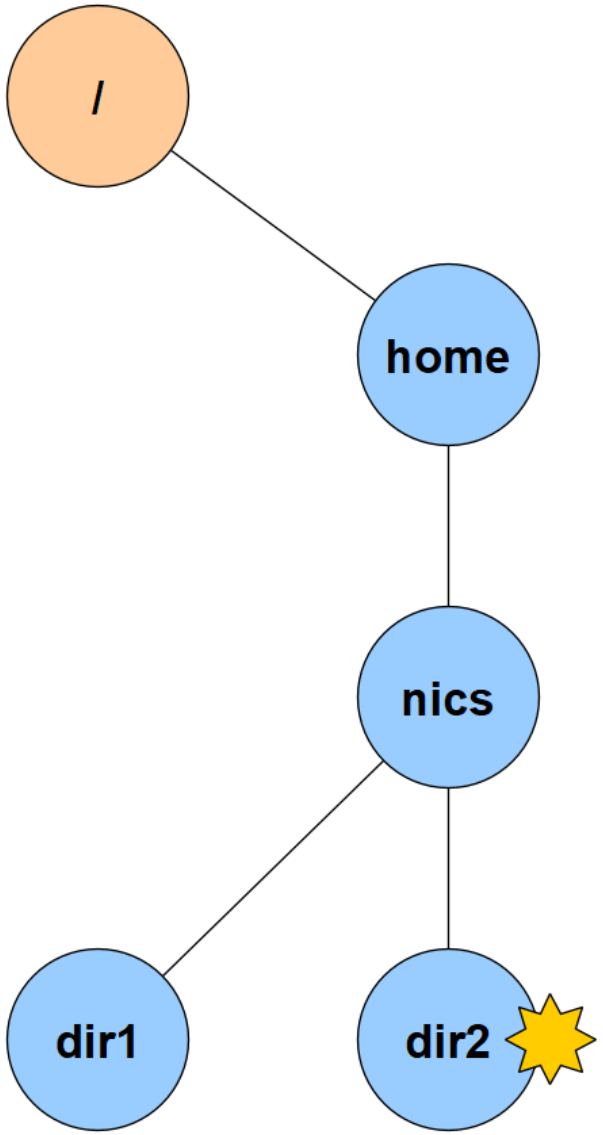
Change Directory

```
nics@linux:~$
nics@linux:~$ cd /home/nics/dir2/
```



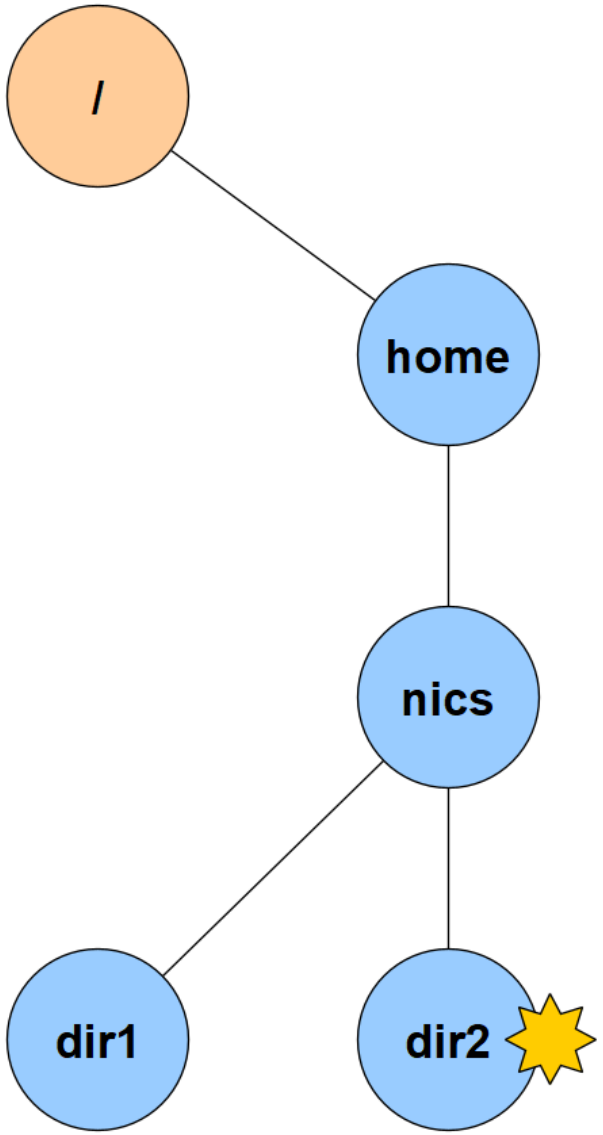
Change Directory

```
nics@linux:~$
nics@linux:~$ cd /home/nics/dir2/
nics@linux:~/dir2$
```



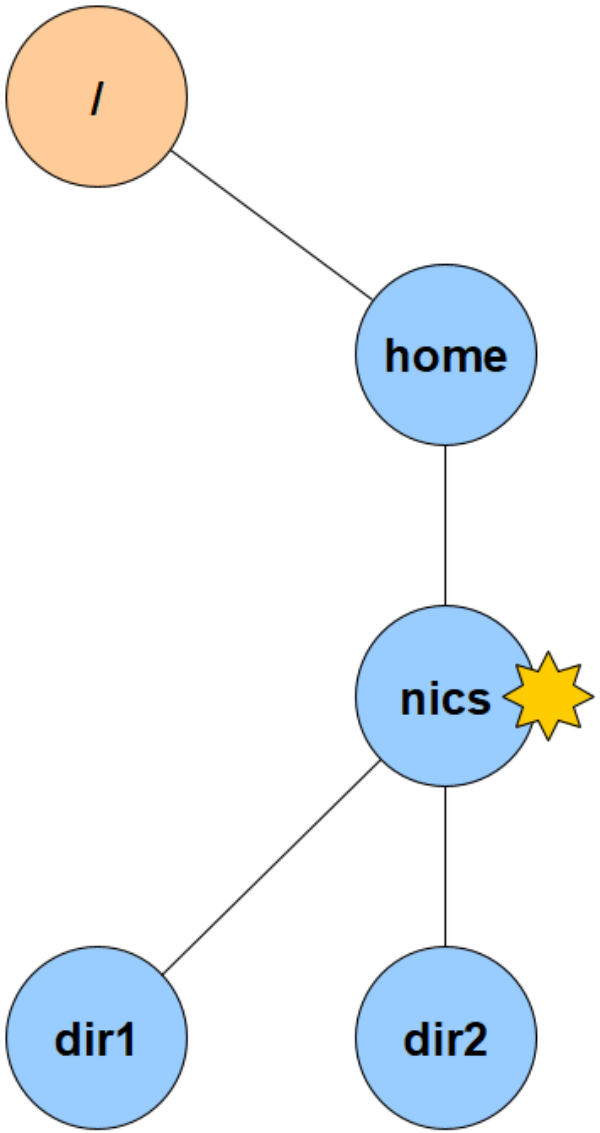
Change Directory

```
nics@linux:~$
nics@linux:~$ cd /home/nics/dir2/
nics@linux:~/dir2$
nics@linux:~/dir2$ cd /home/nics/
```



Change Directory

```
nics@linux:~$
nics@linux:~$ cd /home/nics/dir2/
nics@linux:~/dir2$
nics@linux:~/dir2$ cd /home/nics/
nics@linux:~$
```



Relativsökväg

Filsökväg

Path

Absolut filsökväg

fullständig filsökväg

```
nics@linux:~$ cd /home/nics/dir2/
```

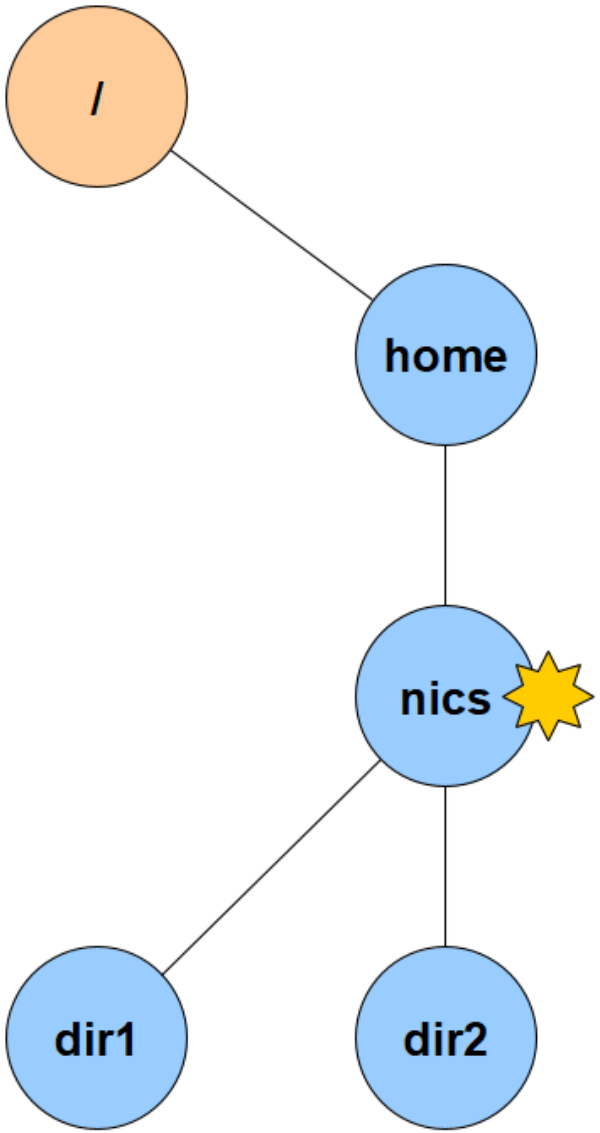
Relativ filsökväg

```
nics@linux:~$ cd dir2/
```

Symbol	Förklaring
.	Nuvarande katalog
..	Överordnad (parent) katalog
~	Användarens hemkatalog
/	Avgränsare i sökvägen, samt roten i filsystemet
\	Escape character Nästkommande tecken behandlas inte som ett special tecken

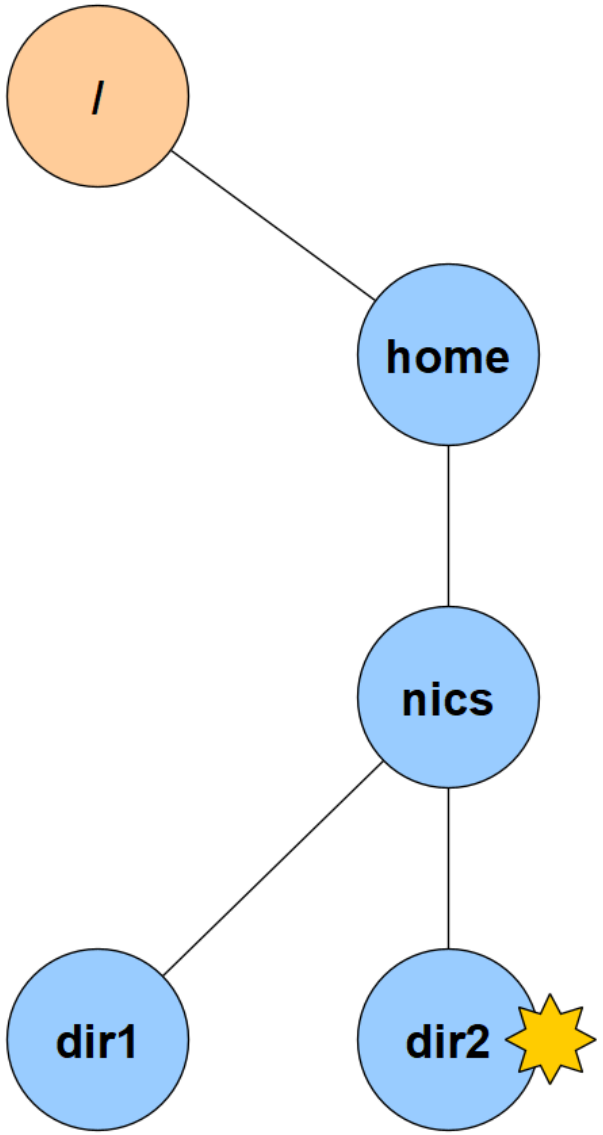
Change Directory

```
nics@linux:~$
nics@linux:~$ cd dir2/
```



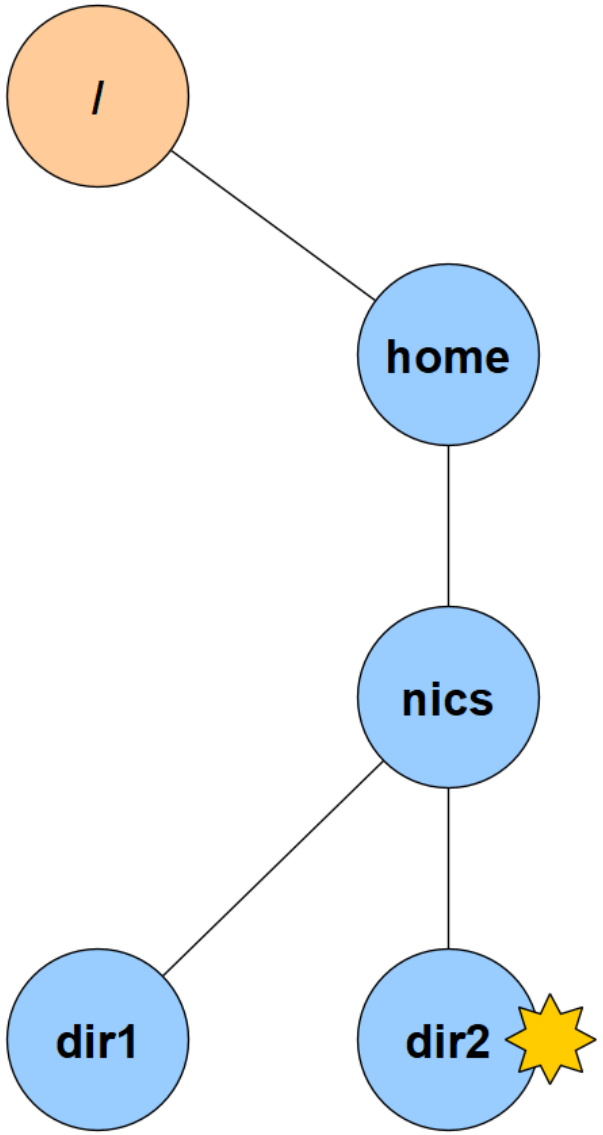
Change Directory

```
nics@linux:~$
nics@linux:~$ cd dir2/
nics@linux:~/dir2$
```



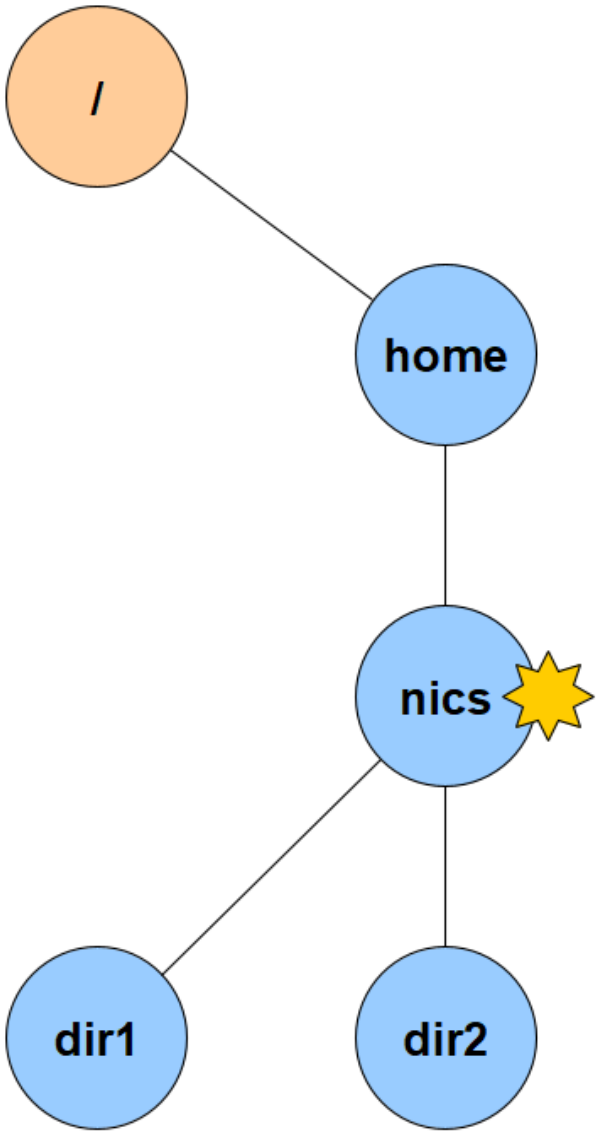
Change Directory

```
nics@linux:~$
nics@linux:~$ cd dir2/
nics@linux:~/dir2$
nics@linux:~/dir2$ cd ..
```



Change Directory

```
nics@linux:~$
nics@linux:~$ cd dir2/
nics@linux:~/dir2$
nics@linux:~/dir2$ cd ..
nics@linux:~$
```



Change Directory

Vad gör detta?

```
nics@linux:~/dir2$ cd ../dir1
```

Hantering av filer & kataloger

List Directory Contents

```
nics@linux:~$ ls
dir1 dir2 file
nics@linux:~$ ls -a
. .. dir1 dir2 file .profile
nics@linux:~$ ls -l
drwxrwxr-x nics nics dir1
drwxrwxr-x nics nics dir2
-rw-rw-r-- nics nics file
```

-a all, do not ignore . or ..

-l use long listing format

Make Directory

```
nics@linux:~$ mkdir demo
nics@linux:~$
nics@linux:~$ ls
demo dir1 dir2 file
```

touch

Skapa en tom fil

```
nics@linux:~$ touch test
nics@linux:~$
nics@linux:~$ ls
demo dir1 dir2 file test
```

Text Editor

Skapa & redigera filer

nano	GNU **nano**
vi	**vi**sual editor
vim	**vi** im**p**roved

```
nics@linux:~$ nano test
```

Print File Contents

cat	print entire file(s) contents
head	print beginning of file(s)
tail	print end of file(s)

```
nics@linux:/var/log$ sudo tail syslog faillog
```

Show File Contents

Om du vill visa hela filinnehållet är en **pager** bättre

more the original pager

less better **more**

```
nics@linux:/var/log$ sudo less syslog
```

Copy

cp copy

cp -r copy everything

```
nics@linux:~$ cp dir1 dir3
nics@linux:~$ cp dir2 dir4
cp: -r not specified;
omitting directory 'dir2'
nics@linux:~$ cp -r dir2 dir4
nics@linux:~$ ls
dir1 dir2 dir3 dir4 file
```

Move

mv move/rename

```
nics@linux:~$ ls
dir1 dir2 dir3 dir4 file
nics@linux:~$ mv dir4 temp
nics@linux:~$ ls
dir1 dir2 dir3 file temp
```

Remove

rm	remove file
rmdir	remove empty folder
rm -r	remove everything

```
nics@linux:~$ rm file
nics@linux:~$ rmdir dir3
nics@linux:~$ rmdir dir2
rmdir: failed to remove 'dir2':
Directory not empty
nics@linux:~$ rm -r dir2
nics@linux:~$ ls
dir1 temp
```

Undo?

Om man råkar göra fel…
  så finns det ingen papperskorg

Tar man bort en fil/katalog så är den borta!

Livelabb

Cheat Sheet

cd	mkdir	cat	cp
pwd	touch	tail	mv
ls	nano	head	rm
	vi/vim	less	rmdir

CAT COMMAND
HAS NOTHING TO DO WITH CATS
quickmeme.com

Utmaning: Navigering

Vad är absoluta sökvägen till **ipxe**?

```
exam@linux:/srv/tftp$ ls
ipxe.efi
exam@linux:/srv/tftp$ ls ../pxe/
ipxe  wimboot  windows10  windows11
exam@linux:/srv/tftp$ ls /home/exam/
exam@linux:/srv/tftp$ cd /etc/
exam@linux:/etc$ cd ssh/
exam@linux:/etc/ssh$ ls
moduli          sshd_config          ssh_host_ecdsa_key.pub
ssh_config      sshd_config.d        ssh_host_ed25519_key
ssh_config.d    ssh_host_ecdsa_key   ssh_host_ed25519_key.pub
exam@linux:/etc/ssh$ cd /etc/nginx/
exam@linux:/etc/nginx$
```

Vad är relativa sökvägen till **sshd_config**?

```
exam@linux:/srv/tftp$ ls
ipxe.efi
exam@linux:/srv/tftp$ ls ../pxe/
ipxe  wimboot  windows10  windows11
exam@linux:/srv/tftp$ ls /home/exam/
exam@linux:/srv/tftp$ cd /etc/
exam@linux:/etc$ cd ssh/
exam@linux:/etc/ssh$ ls
moduli          sshd_config          ssh_host_ecdsa_key.pub
ssh_config      sshd_config.d        ssh_host_ed25519_key
ssh_config.d    ssh_host_ecdsa_key   ssh_host_ed25519_key.pub
exam@linux:/etc/ssh$ cd /etc/nginx/
exam@linux:/etc/nginx$
```

Användare & filrättigheter

Superuser

root Har obehindrad behörighet till alla kommandon & filer

root-kontot ska man dock undvika att använda

sudo Kör ett kommando som root

sudoedit Redigera en fil som root

```
nics@linux:~$ sudo mkdir /srv/files
nics@linux:~$ sudoedit /etc/hosts
```

UNIX File Permissions

Mycket enklare filrättigheter jämfört med t.ex. NTFS

Standardiserade i POSIX

Rättigheter tilldelas: ACL Subjects

- User owner
- Group owner
- Other (everyone else)

UNIX File Permissions

Ägaren kan förändra de rättigheter som är satta på ett objekt (fil/katalog)

Finns alltid två ägare

User owner **Group owner**

Hur gör man om man bara vill att en användare ska få behörighet?

UNIX File Permissions

```
nics@linux:~$ ls -l
-rw-r--r--   nics demo file
drwxr-xr-x   nics nics folder
               ^     ^
               |     |
Owner:       user  group
```

UNIX File Permissions

ACL Permission	File	Directory
Read	Se innehåll i filer	Se innehåll i kataloger
Write	Förändra filers innehåll	Förändra katalogers innehåll (skapa, döp om, ta bort filer/ kataloger)
E**x**ecute	Kör filer (i.e. script)	Navigera till katalogen (i.e. **cd**)

UNIX File Permissions

```
nics@linux:~$ ls -l
-rw-r--r--    <output omitted>
drwxr-xr-x    <output omitted>
 ^^^
 |||
r = read
w = write
x = execute
```

UNIX File Permissions

```
nics@linux:~$ ls -l
-rw-r--r--   nics demo file
drwxr-xr-x   nics nics folder
```

User	Group	Other
R W X	R W X	R W X

Change File Mode Bits

```
nics@linux:~$ chmod 700 folder
nics@linux:~$ ls -l
<output omitted>
drwx------   nics nics folder
```

User			Group			Other		
R	W	X	R	W	X	R	W	X
4	2	1	4	2	1	4	2	1

Övning

```
r--r----- = 440

r-xr--r-- = 544

rwxr-xr-x = 755
```

User			Group			Other		
R	W	X	R	W	X	R	W	X
4	2	1	4	2	1	4	2	1

Change File User Ownership

```
nics@linux:~$ ls -l
-rwxrwxrwx   nics nics file
<output omitted>
nics@linux:~$ chown demo file
nics@linux:~$ ls -l
-rwxrwxrwx   demo nics file
<output omitted>
```

Change File Group Ownership

```
nics@linux:~$ ls -l
-rwxrwxrwx   demo nics file
<output omitted>
nics@linux:~$ chgrp demo file
nics@linux:~$ ls -l
-rwxrwxrwx   demo demo file
<output omitted>
```

Tips

Manual Pages

RTFM!

man show manual

man x show manual page x

```
nics@linux:~$ man rmdir
<output omitted>

SEE ALSO
       rmdir(2)

<output omitted>
nics@linux:~$ man 2 rmdir
```

Manual Pages

```
nics@linux:~$ man man
<output omitted>
The table below shows the section numbers of the manual followed by the types of
pages they contain.

       1   Executable programs or shell commands
       2   System calls (functions provided by the kernel)
       3   Library calls (functions within program libraries)
       4   Special files (usually found in /dev)
       5   File formats and conventions, e.g. /etc/passwd
       6   Games
       7   Miscellaneous (including macro packages and conventions), <output omitted>
       8   System administration commands (usually only for root)
       9   Kernel routines [Non standard]

       A manual page consists of several sections.
<output omitted>
```

Apropos

apropos search manual pages

```
nics@linux:~$ apropos rmdir
rmdir (1)    - remove empty directories
rmdir (2)    - delete a directory
```

Help

--help simple how-to-use information

```
nics@linux:~$ rmdir --help
Usage: rmdir [OPTION]... DIRECTORY...
Remove the DIRECTORY(ies), if they are empty.

<output omitted>
```

Tabb

[tabb]	auto-complete file path
[tabb][tabb]	suggestions

```
nics@linux:~$ ls
demo dir3 temp
nics@linux:~$ cd d[tabb][tabb]
demo dir3
nics@linux:~$ cd di[tabb]
nics@linux:~$ cd dir3
```

History

history	print history
!x	run number x from history
!!	run last command in history

```
nics@linux:~$ history
<output omitted>
 6  cd dir3
nics@linux:~$ !!
cd dir3
nics@linux:~/dir3$
```

Reverse Search

^R CTRL+R, search for command in history

```
nics@linux:~$ ^R
(reverse-i-search)`cd': cd dir3
```

Avbryt & avsluta

^C CTRL+C, avbryt

Q Avslutar många program

```
nics@linux:~$ ls -R /
<output omitted>
^C
nics@linux:~$ less /etc/network/interfaces
<output omitted>
q
```

The ABCs of UNIX

A is for `awk`, which runs like a snail, and
B is for `biff`, which reads all your mail.

C is for `cc`, as hackers recall, while
D is for `dd`, the command that does all.

E is for `emacs`, which rebinds your keys, and
F is for `fsck`, which rebuilds your trees.

G is for `grep`, a clever detective, while
H is for `halt`, which may seem defective.

I is for `indent`, which rarely amuses, and
J is for `join`, which nobody uses.

K is for `kill`, which makes you the boss, while
L is for `lex`, which is missing from DOS.

M is for `more`, from which `less` was begot, and
N is for `nice`, which it really is not.

O is for `od`, which prints out things nice, while
P is for `passwd`, which reads in strings twice.

Q is for `quota`, a Berkeley-type fable, and
R is for `ranlib`, for sorting a table.

S is for `spell`, which attempts to belittle, while
T is for `true`, which does very little.

U is for `uniq`, which is used after `sort`, and
V is for `vi`, which is hard to abort.

W is for `whoami`, which tells you your name, while
X is, well, `X`, of dubious fame.

Y is for `yes`, which makes an impression, and
Z is for `zcat`, which handles compression.

Till nästa gång

- Läs kurslitteraturen
- **Gå med i labbgrupp!**
- Onsdag/torsdag: labb

Tack för idag